

Data Collection and Preprocessing Phase

Date	01 August 2026
Team ID	_____
Project Title	AI Adoption and Impact in Global Education
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	The project uses a global AI-in-education dataset containing variables related to student and teacher AI usage, school adoption, daily AI usage time, internet penetration, education index, government AI policies, AI tools, countries/regions, and demographics. The dataset will be explored to understand AI adoption patterns, regional differences, and factors associated with AI integration in education.
Data Cleaning	Inspect the dataset for missing values, duplicate records, inconsistent entries, and invalid or incorrect values. Remove exact duplicates, correct identifiable data-entry errors, and handle missing values using an appropriate method based on the affected field. Records that cannot be reliably corrected will be reviewed before analysis.
Data Transformation	Filter and sort records where required for analysis. Standardize categorical values such as country, region, demographic group, and AI tool. Create analysis-ready calculated fields where required, such as adoption comparisons, usage summaries, and aggregated indicators for Tableau dashboards.
Data Type Conversion	Check and rectify data types for every field. Convert numeric measures such as adoption rates, usage time, internet penetration, and education indicators to appropriate numeric types. Ensure categorical fields are stored consistently and any date/time fields are converted to suitable date or time formats if present.
Column Splitting and Merging	Split columns when a single field contains multiple logical values, such as combined location information, if required by the dataset. Merge or derive fields when separate columns need to be combined for analysis. Only perform these operations where the actual dataset structure requires them.
Data Modeling	Prepare the cleaned dataset for Tableau analysis. Define relationships between tables if multiple related tables are present and create calculated measures required for the dashboards. Organize dimensions and measures so that country, region, demographic, adoption, usage, infrastructure, policy, and AI-tool analysis can be performed efficiently.
Save Processed Data	Save the cleaned and transformed dataset in a structured format for future analysis and Tableau dashboard development. Maintain a separate processed version so that the original dataset remains unchanged. The final processed data will be used as the source for the project's exploratory analysis and interactive

	dashboards.
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Note: Exact dataset size, file format, missing-value counts, duplicate counts, and specific data-type corrections should be recorded after the actual dataset is inspected.